

Video Transcript

Video 11: OpenSSL: PKI Keys, Encryption, and Signing (09:24)

So, let's take a look at how to use OpenSSL. The first thing we need to do is to do a brew install. So, let's take a look at OpenSSL version, just to check that we've got it and we're running 2.8.3. So, let's take a look at how we use keys. So, we're going to generate some OpenSSL keys and we're going to show first how to sign a document, and then we'll look at how to encrypt a document and also sign it. So first, we need to generate a key. And to do this, we're going to use OpenSSL and what I need to do is to first generate a key.

So, let's generate a key. So, we use OpenSSL genrsa. We're going to use an RSA key of 512 bits and we're going to put it to a file called called 'myprivate.pem'. So, this is going to use a standard called pem. We'll see later that we may have keys in using a different standard, but this one generates a key used called 'myprivate. pem'. Let me just cat that show you. So, that's what it looks like. Begin RSA private key and then end RSA private key.

Okay so, that's our private key. Now, what we need to do, let me clear, we need to generate from that our public key. We're going to use our private key to output our public key. So, let's take a look. Now, we should have our public key. So, here's our public key and you can see it's typically shorter than the private key. So, now we have our two keys. Let's sign a document. So, the document we're going to sign, I've already got is a file called hello.txt. So, it just says hello world. So, we're just going to sign that file.

Notice this one is not encrypted and basically what we're going to do if we go back to our diagram. So, what we're going to do now is take this file and we're going to create a hash code that needs to be padded. But then we're going to use our private key to encrypt that hash code. And so that's called our signature. Now, the only way to decrypt is with our public key, which is widely shared. So, what we're going to do then is we're going to share the file with another person and we're going to share our signature. And they've already got our public key, which is widely distributed.

So, they apply the public key to the signature, decrypt it, they get the padded hash, they then get the unpadded hash. Now, because I sent them the file, they can take the hash of that, and they can compare it with the hash they've just decrypted. And if those two match, it means that I must have signed with my private key because my public key generates or decrypts and gets the same hash. And so, you know that the file came from me. So, that's what we're going to do now. So, let's go back to our terminal.

So, now we're going to use OpenSSL and notice we're taking the digest and that's going to be a sha1 digest. And we're going to sign my with my private key and we're going to output to sha1 sign. And we're going to sign the file hello.txt, okay? So, let's do that. And if I cat sha1.sign, that's our signature. So, now we send our signature and the hello.txt file to, for example, our friend. They can access my public signature. So, they've got all three of those. Now, what do they do? Well, they want to verify that it came from me.

So, they now want to verify using my public key and the signature that it was applied to this document, hello.txt. So, they have those three pieces of information. Now, they're going to do exactly what I said and see if it came from me and they verify that it does. They used my public key to decrypt. So, just to remind you, this is what they did. So, we signed with our private key, which basically signed the hash of the document. They encrypted the hash and that's our signature. So, we sent the document separately and we sent this hash and our signature basically, and they used our public key to decrypt it.

They took the hash of the document and compared it with the hash that they got by decryption. And if those hashes matches, they know the file came from me. So, now let's see how this was a file sent in the open 'Hello world.' But, let's suppose now we want to encrypt that. So, what we're going to do now is encrypt it. And so, our file now will be this file that's inputted will be an encrypted file, but we're just going to sign it in the same way. So, now that allows us to send hidden text to our friend, so we'll go back and we'll encrypt the document. So, let's take a look now at encrypting the hello.txt.

So, now we need to encrypt with the public key of my friend. So, I've generated my friends keys and here's their public key. And now what I need to do is I'm going to use their public key here to encrypt the hello.txt and I want the output put into encrypted hello. So, let's do that. And let's look at encrypted hello. So, to decrypt that encrypted message, we need to use the private key.

This is my friend's private key. So, this is my friend doing it. And they've got the encrypted text and now they can decrypt.

So, let's do that and they're going to put it in my decrypted. And we've retrieved 'Hello world.' So, that's how we encrypt a message. Now, what we need to do, we actually would sign, we'd go through that signature process on this encrypted message. So, the instead of feeding the plain text 'Hello world' in, we would now sign this encrypted hello. So, that's how signatures work. There are some limitations in the size of file that we can encrypt this way, but essentially that's how you encrypt and decrypt documents and how you sign them. So, bye for now.

